PHD125 Pharmaceutical Organic Chemistry · Lipids

Before class: the molecules that hate water

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Read the cards first, then answer the questions without notes. The answer key is at the back.

1 · Read first (5 min)

Defined by what they avoid

Lipids are not one functional group. They are the molecules in living things that dissolve in oily, nonpolar solvents but not in water: fats and oils, waxes, the phospholipids of cell membranes, and steroids such as cholesterol.

Esters again

Most lipids are **esters** (Chapter 4). A fat is glycerol (a triol, Chapter 3) joined to three long-chain carboxylic acids called **fatty acids**. Boil a fat with NaOH and the esters split: you get glycerol and soap. That is saponification, which you met in Chapter 4.

Kinks and walls

A cis C=C (Chapter 2) puts a kink in a fatty acid chain, so oils are liquid and butter is solid. Phospholipids have a water-loving head and two water-fearing tails; in water they line up into a double layer, the **cell membrane**. Every drug you swallow has to cross it.

2 · Readiness check

Ten quick questions: recap and a preview.

Q1 Recap

Glycerol, $HOCH_2CH(OH)CH_2OH$, is a:

A. triol (alcohol with three –OH groups)
B. carboxylic acid
C. ester
D. ketone

Q2 Recap

Basic hydrolysis of an ester with NaOH gives:

A. a carboxylate salt and an alcohol
B. an amide
C. an alkene
D. a ketone

Q3 Recap

Adding H_2 (Ni catalyst) to C=C bonds is called:

A. hydrogenation
B. hydration
C. hydrolysis
D. halogenation

Q4 CH6-LO1

Lipids are best described as compounds that:

A. dissolve in nonpolar solvents but not in water
B. contain nitrogen
C. are always solids
D. are sugars

Q5 CH6-LO1

Fats provide more energy per gram than carbohydrates.

True / False

Q6 CH6-LO2

Why is olive oil liquid but butter solid?

A. Olive oil has more unsaturated (kinked) fatty acids
B. Olive oil has no glycerol
C. Butter is a wax
D. Butter contains water

Q7 CH6-LO1

Match each lipid to its class.

Options: triacylglycerol · wax · steroid · phospholipid

palm oil _____

beeswax _____

cholesterol _____

lecithin _____

Q8 CH6-LO4

Cell membranes are mainly made of:

A. a double layer of phospholipids
B. cellulose
C. triglycerides
D. waxes

Q9 CH6-LO5

Hydrocortisone cream contains a:

A. steroid
B. wax only
C. phospholipid
D. sugar

Q10 CH6-LO3

Why does an opened bottle of cooking oil eventually smell bad, even with the cap on?

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Answer key

2 · Readiness check

Q1 Recap

Glycerol, $HOCH_2CH(OH)CH_2OH$, is a:

A. triol (alcohol with three –OH groups)

B. carboxylic acid

C. ester

D. ketone

Chapter 3.

Q2 Recap

Basic hydrolysis of an ester with NaOH gives:

A. a carboxylate salt and an alcohol

B. an amide

C. an alkene

D. a ketone

Saponification (Chapter 4).

Q3 Recap

Adding H_2 (Ni catalyst) to C=C bonds is called:

A. hydrogenation

B. hydration

C. hydrolysis

D. halogenation

Chapter 2 addition reactions.

Q4 CH6-LO1

Lipids are best described as compounds that:

A. dissolve in nonpolar solvents but not in water

B. contain nitrogen

C. are always solids

D. are sugars

They are defined by solubility.

Q5 CH6-LO1

Fats provide more energy per gram than carbohydrates.

True / False → **True**

About 9 kcal/g versus 4 kcal/g.

Q6 CH6-LO2

Why is olive oil liquid but butter solid?

A. Olive oil has more unsaturated (kinked) fatty acids

B. Olive oil has no glycerol

C. Butter is a wax

D. Butter contains water

Kinked chains pack poorly.

Q7 CH6-LO1

Match each lipid to its class.

palm oil → triacylglycerol; beeswax → wax; cholesterol → steroid; lecithin → phospholipid

Four of the classes in this chapter.

Q8 CH6-LO4

Cell membranes are mainly made of:

A. a double layer of phospholipids

B. cellulose

C. triglycerides

D. waxes

The lipid bilayer.

Q9 CH6-LO5

Hydrocortisone cream contains a:

A. steroid

B. wax only

C. phospholipid

D. sugar

A corticosteroid.

Q10 CH6-LO3

Why does an opened bottle of cooking oil eventually smell bad, even with the cap on?

Oxygen in the air (and light and warmth) slowly attacks the C=C bonds in the unsaturated fatty acids, breaking them into small smelly aldehydes and acids: the oil turns rancid. Moisture can also hydrolyse the esters, releasing smelly free fatty acids.

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